

OOP Foundations Demo Syllabus

Polymorphism allows one interface to support many concrete behaviors in object-oriented programming. The syllabus defines runtime polymorphism as methods resolved through inheritance.

Inheritance establishes the common parent-child relationship used to enable polymorphic method dispatch in class hierarchies.

Figure 1 shows the class hierarchy architecture for polymorphism, method overriding, and dynamic binding.

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Method overloading and method overriding are distinct. Overloading changes parameter list while overriding replaces inherited behavior in a subclass.

Example: a drawing system can call draw on many shapes, and each subclass has its own implementation.

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Abstraction and encapsulation usually appear with polymorphism because they are grouped together as core object oriented principles.

The review section emphasizes related concepts such as dynamic binding, inheritance, and interface design.